KAIZEN MATHS

Lesson starter

Large Data Set Practice

Exam Interpretation

Name _____

S1

State one limitation of the Kaizen Large Data Set and explain how it might affect a conclusion.

S2

Exam-style: A student uses the model sunshine = 210 - 1.55(rainfall) from the Weather & Climate Data and claims it can be used for every similar situation. Give two reasons why this claim may be unreasonable.

S3

Exam-style: A student uses the model rainfall = 18.6 + 2.15(wind speed) from the Weather & Climate Data and claims it can be used for every similar situation. Give two reasons why this claim may be unreasonable.

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Teacher answers & working

Large Data Set Practice · Exam Interpretation

STA-986D-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

State one limitation of the Kaizen Large Data Set and explain how it might affect a conclusion.

Answer:

Correlation between weather variables does not prove that one causes the other.

Working:

Choose a limitation linked to the context.

Correlation between weather variables does not prove that one causes the other.

Explain the effect: the conclusion may be less reliable, less precise, or not generalisable.

S2 / E1

Exam-style: A student uses the model sunshine = 210 - 1.55(rainfall) from the Weather & Climate Data and claims it can be used for every similar situation. Give two reasons why this claim may be unreasonable.

Answer:

Limited data and extrapolation/causation risk

Working:

Reason 1: the data set is a limited training dataset, so it may not represent every possible situation.

Reason 2: the model should only be used within the range of observed data; outside that range would be extrapolation.

Reason 3: even with correlation, the model does not prove a causal relationship.

A strong answer links each reason to the context of the data.

S3 / E2

Exam-style: A student uses the model rainfall = 18.6 + 2.15(wind speed) from the Weather & Climate Data and claims it can be used for every similar situation. Give two reasons why this claim may be unreasonable.

Answer:

Limited data and extrapolation/causation risk

Working:

Reason 1: the data set is a limited training dataset, so it may not represent every possible situation.

Reason 2: the model should only be used within the range of observed data; outside that range would be extrapolation.

Reason 3: even with correlation, the model does not prove a causal relationship.

A strong answer links each reason to the context of the data.